Research Report

Iot Application in smart Agriculture

Submitted To

Code Alpha

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Abstract

Agriculture plays an important role in our daily lives, but traditional farming often requires farmers to continuously monitor conditions such as temperature, rainfall, soil moisture, and humidity. To make this process easier and more efficient, this project presents an IoT-Based Smart Farming System that can monitor these conditions and automatically control basic farming operations.

The system uses an Arduino Uno as the main controller. Sensors are used to measure temperature, detect rain, and monitor soil moisture and humidity. The collected information is displayed on a 20×4 LCD, allowing the user to easily observe the current conditions. The system also controls a fan and water pump through relay modules. When the temperature rises above a certain level, the fan is automatically switched ON. Similarly, when the soil becomes dry and there is no rain, the water pump automatically starts to provide irrigation. When the required conditions are achieved, the pump is switched OFF.

Introduction

The IoT-Based Smart Farming System is designed to make farming easier through automatic monitoring and control. The system uses Arduino Uno with sensors to monitor temperature, rain, soil moisture, and humidity.

The sensor values are displayed on a 20×4 LCD. If the temperature becomes high, the fan automatically turns ON. When the soil becomes dry and there is no rain, the water pump automatically starts for irrigation.

This project helps reduce manual effort, save water, and improve farming efficiency. The complete system is also tested using Proteus simulation.

Objective

The main objectives of the IoT-Based Smart Farming System are:

1. To monitor temperature, rainfall, soil moisture, and humidity automatically.
2. To display the sensor readings on a 20×4 LCD for easy monitoring.
3. To automatically turn ON the fan when the temperature goes above 30°C.
4. To automatically turn ON the water pump when soil moisture falls below 40% and there is no rain.
5. To reduce manual effort in monitoring and irrigation.
6. To save water by providing irrigation only when required.

7.To demonstrate the use of Arduino, sensors, relays, and automation in smart agriculture.

8.To test and verify the complete system using Proteus simulation

Hardware & Software Requirements

. Hardware Requirements

1. Arduino Uno – Acts as the main controller and processes all sensor readings.
2. LM35 Temperature Sensor – Measures the surrounding temperature.
3. Rain Sensor – Detects whether rain is present or not.
4. Soil Moisture Sensor – Measures the moisture level of the soil.
5. Humidity Sensor – Measures the humidity level of the environment.
6. 20×4 LCD Display – Displays temperature, rain status, soil moisture, humidity, and actuator status.
7. Relay Modules – Used to control the fan and water pump.
8. 12V DC Fan – Automatically operates when the temperature is high.
9. 12V Water Pump – Provides automatic irrigation when the soil is dry.
10. 12V Batteries/Power Supply – Provides power to the fan and water pump.
11. Connecting Wires – Used for electrical connections between the components.

. Software Requirements

1. LCD Initialization:

The LCD is initialized as a 20×4 display and the project title "IoT SMART FARMING" is displayed.

2. Temperature Monitoring:

The Arduino reads the LM35 through analog pin A0. The analog value is converted into temperature in °C.

If temperature is above 30°C, the fan turns ON.

Otherwise, the fan remains OFF.

3. Rain Detection:

The rain sensor is connected to A1 and provides a digital signal.

R = 0 → No rain

R = 1 → Rain detected

4. Soil Moisture Monitoring:

The soil moisture value is read through A2 and converted into a percentage for display. If the moisture level falls below 40%, the soil is considered dry.

5. Humidity Monitoring:

The humidity input is connected to A3. Its value is converted into a percentage and displayed on the LCD.

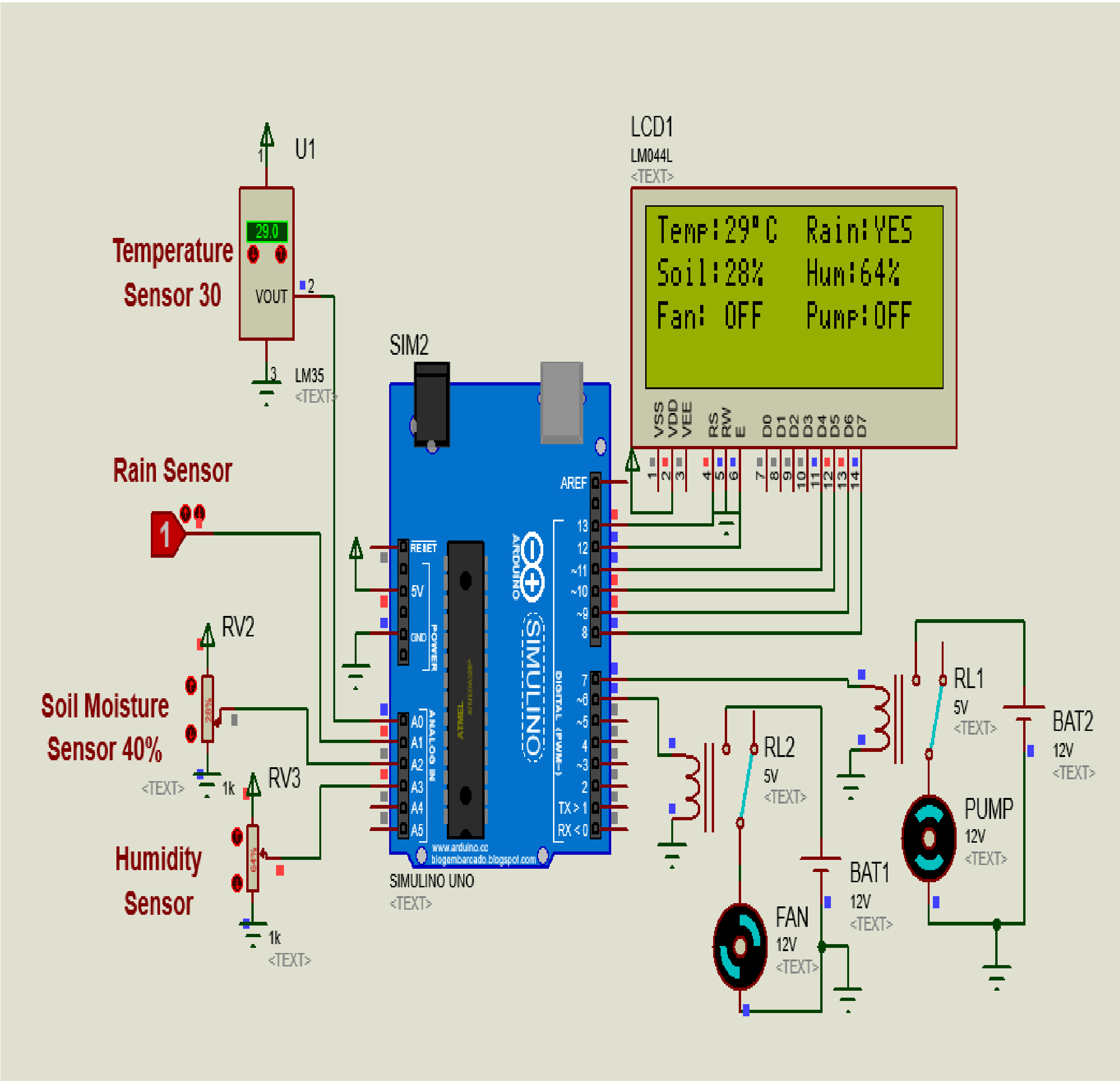
6. Automatic Irrigation:

The pump operates only when both conditions are satisfied:

Soil moisture is below 40%

No rain is detected

When these conditions are true, the pump relay is activated and the water pump turns ON.



Working Principle

The IoT-Based Smart Farming System works by continuously monitoring the environmental conditions of the farm and automatically controlling the fan and water pump. Arduino Uno acts as the main controller and processes the data received from different sensors.

First, the LM35 temperature sensor measures the surrounding temperature and sends the analog value to Arduino through A0. If the temperature rises above 30°C, Arduino activates the relay connected to the fan, and the fan turns ON. When the temperature is 30°C or below, the fan remains OFF.

The rain sensor is connected to A1 and detects whether rain is present. At the same time, the soil moisture sensor connected to A2 measures the moisture level of the soil. If the soil moisture falls below 40% and no rain is detected, Arduino activates the pump relay and the water pump turns ON. If the soil has sufficient moisture or rain is detected, the pump remains OFF.

The humidity sensor connected to A3 provides the humidity value, which is displayed on the 20×4 LCD along with temperature, rain status, soil moisture, and the ON/OFF status of the fan and pump.

Thus, the system continuously follows this cycle:

Sensors → Arduino → Data Processing → LCD Display → Fan/Pump Control

Proteus Simulation & Result

The complete smart farming circuit was designed and simulated in Proteus before practical hardware implementation. Arduino Uno was connected with the LM35 temperature sensor, rain sensor, soil moisture sensor, humidity input, 20×4 LCD, relay modules, fan, and water pump.

During the simulation, different sensor values were provided to check the system response. The LCD displayed the sensor readings and the status of the fan and pump. When the temperature exceeded 30°C, the fan was activated. When the soil moisture was below 40% and no rain was detected, the water pump was activated. In other conditions, the respective devices remained OFF.

Result

The Proteus simulation successfully verified the working of the circuit and the programmed control conditions. The LCD displayed the required sensor information, and the fan and water pump responded automatically according to the defined conditions. Thus, the simulation confirmed that the proposed smart farming system works as intended.

Future Scope

The proposed smart farming system can be further improved by adding IoT and advanced automation features. In the future, the system can be connected to the Internet through Wi-Fi or GSM so that farmers can monitor farm conditions remotely using a mobile application.

Other sensors such as water level, soil pH, light intensity, and nutrient sensors can also be added for better crop monitoring. Weather forecast data can be integrated to avoid unnecessary irrigation when rain is expected.

The collected sensor data can also be stored in the cloud for analysis and monitoring. In the future, AI and machine learning can be used to predict irrigation requirements and provide smarter decisions based on crop and environmental conditions.

Conclusion

The IoT-Based Smart Farming System successfully demonstrates the use of Arduino, sensors, LCD, and relay-controlled devices for basic agricultural automation. The system monitors temperature, rain, soil moisture, and humidity and automatically controls the fan and water pump according to predefined conditions.

The complete circuit was tested in Proteus simulation, where the fan and pump responded correctly to the sensor conditions. Overall, the project helps reduce manual effort, save water, and improve the efficiency of farming operations. It also provides a good foundation for developing more advanced smart farming systems using IoT, cloud monitoring, and AI technologies.